# Ali Kamran

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## PROFESSIONAL SUMMARY

Computer Science student specializing in embedded systems and low-level computing, with research experience in multicore computer architecture and timing predictability for safety-critical systems. Skilled in developing real-time applications on STM32 microcontrollers using FreeRTOS, integrating u-blox GPS modules via UART, and transmitting data over CAN bus. Proficient in C/C++ and Python in Linux environments, with hands-on experience in robotics, sensor integration, and real-time control systems.

## EDUCATION

**Honours Bachelor of Computer Science** 2024 – 2027
York University, Toronto, Ontario
Relevant Coursework:

- ***Embedded Systems, Digital Logic Design, Computer Architecture(RISC-V, System Verilog),Computer Organization, Operating System Fundamentals, Signals and Systems , Design & Analysis of Algorithms***

## EXPERIENCE

**Capstone Project - Timing-Predictable Atomic Synchronization for Safety-Critical CPS**
Supervisor: Anirudh M. Kaushik January 2026 – Present
YorkArch Research Lab, York University Toronto, Ontario

**Embedded Member** February 2025 – August 2025
York University Robotics Society, Toronto, Ontario

- Built a GPS-based navigation node in ROS2 Humble using Python to calculate distance and heading between coordinates.
- Programmed STM32F3026T6 using FreeRTOS to interface with a u-blox Neo-M8N GPS via UART and transmit location data over CAN bus to a Jetson module for a Mars rover navigation system.
- Implemented ADC-based USB voltage monitoring on STM32F3026T6 ARM Cortex-M4 and broadcasted readings over CAN bus.

## PERSONAL PROJECTS

**Object Detection Using ROS2 & YOLOv8**

- Developed a ROS 2-based computer vision pipeline integrating camera nodes for real-time image capture and YOLOv8 for object detection and classification, enabling efficient processing and communication between sensor and compute modules.

**STM32 Air Quality Monitor (ENS160 + AHT21)**

- Designed and implemented an STM32-based air quality monitoring system using I²C to acquire environmental data from AHT21 (temperature, humidity) and send compensation inputs to the ENS160 gas sensor, outputting real-time $eCO_2$, TVOC, and AQI readings via UART.

**FPGA Note Generator (DE10-Lite) — Verilog, UART, C++ (PortAudio)**

- Designed and implemented an FPGA-based selectable note generator on a DE10-Lite using Verilog RTL, including on-chip frequency measurement (rising-edge timing), 7-segment display output, and UART (115200 8N1) telemetry to a host program for real-time audio playback.

## TECHNICAL SKILLS

- **Embedded Systems:** ARM Cortex-M4, ATmega328P
- **Programming Languages:** C, C++, Python, C#, Java, Verilog, RISC-V assembly, Bash, PowerShell, JavaScript
- **Hardware & Prototyping:** electronic prototyping, soldering
- **Test & Measurement:** oscilloscopes, multimeters, spectrum analyzers, VNAs, logic analyzers
- **Tools & Platforms:** Git, Linux (Ubuntu), STM32CubeIDE, VS Code, Make, CMake, Windows
- **Office & Collaboration:** Microsoft Office (Excel), Teams, Google Drive
- **Databases:** MySQL, PostgreSQL, SQLite, Microsoft SQL Server, MongoDB